

Python Programming

Unit 04 – Lecture 01: Tkinter Introduction and Widgets

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Repository: <https://github.com/tali7c/Python-Programming>

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Learning Outcomes

- Create a basic Tkinter window and run the GUI loop

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- Understand widgets and how they are placed in a window
- Use common widgets: Label, Button, Entry, Text, Canvas, Listbox
- Use selection widgets: Checkbutton, Radiobutton, Combobox, Spinbox

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- Works with widgets (buttons, text boxes, etc.)

Smallest Tkinter App

```
import tkinter as tk

root = tk.Tk()
root.title("My App")
root.geometry("400x200")
root.mainloop()
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- mainloop() keeps the window running

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 - configuration options (text, width, command, ...)

Label, Button, Entry (Example)

```
import tkinter as tk

root = tk.Tk()
name_var = tk.StringVar()

tk.Label(root, text="Name: ").pack()
tk.Entry(root, textvariable=name_var).pack()

def greet():
    print("Hello, ", name_var.get())

tk.Button(root, text="Greet", command=greet).pack()
root.mainloop()
```

More Widgets

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- Canvas: drawing shapes
- Scrollbar: scrolling support (Text/Listbox)
- Listbox: list selection

Selection Widgets

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Demo: Widget Showcase

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 - ttk Combobox

Checkpoint 1

Question: Why do we need `root.mainloop()` in Tkinter?

Checkpoint 2

Question: What is the benefit of using `StringVar` with an `Entry` widget?

Think-Pair-Share

Discuss:

- What makes a GUI *easy to use*?
- Give 3 design rules (labels, spacing, validation, feedback, etc.).

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- Tk() creates a window; mainloop() runs the event loop
- Widgets like Label/Button/Entry handle input and actions
- Selection widgets (check/radio/combobox) capture choices

Exit Question

Write the 3 key lines needed to create a Tkinter window titled "Hello GUI" and keep it running.